Sols

Goal: To collect information about the first and second derivatives of a function, then use this information to graph the function without using technology.

1. Consider the function $f(x) = 3x^4 - 8x^3 + 6x^2$.

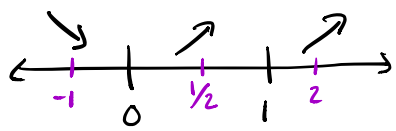
   (a) Determine the open intervals on which the function is increasing/decreasing.

$f'(x) = 12x^3 - 24x^2 + 12x$

$= 12x(x^2 - 2x + 1)$

$= 12x(x-1)^2$

$f'(x) = 0$ when $x = 0$, $x = 1$



$f'(-1) < 0$ $f'(1/2) > 0$ $f'(2) > 0$

Increasing on $(0,1) \cup (1,\infty)$

Decreasing on $(-\infty, 0)$

   (b) Find the local maxima and local minima of $f(x)$, if any. Be sure to find the critical points, classify them using either the first or second derivative test, then substitute the x-values into $f(x)$ to find the local mininum/maximum values.

From part (a) the critical points occur at $x=0$, $x=1$. By the First Derivative Test, there is a local minimum at $x=0$ or $(0, f(0)) = (0,0)$.

   (c) Find the inflection points of the function, if any. Be sure to find where the second derivative is zero, use a sign chart to determine whether or not the second derivative changes, then substitute the x-values into $f(x)$ to find the y-value at each inflection point.

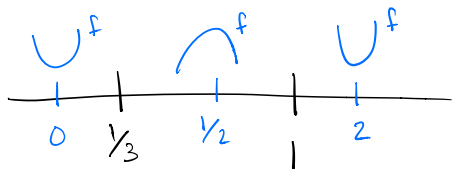
$f''(x) = 36x^2 - 48x + 12$

$36x^2 - 48x + 12 = 0$

$12(3x^2 - 4x + 1) = 0$

$12(3x-1)(x-1) = 0$

$x = 1/3$, $x = 1$



$f''(0) = 12 > 0$

$f''(1/2) = 12(3/2 - 1)(1/2 - 1) < 0$

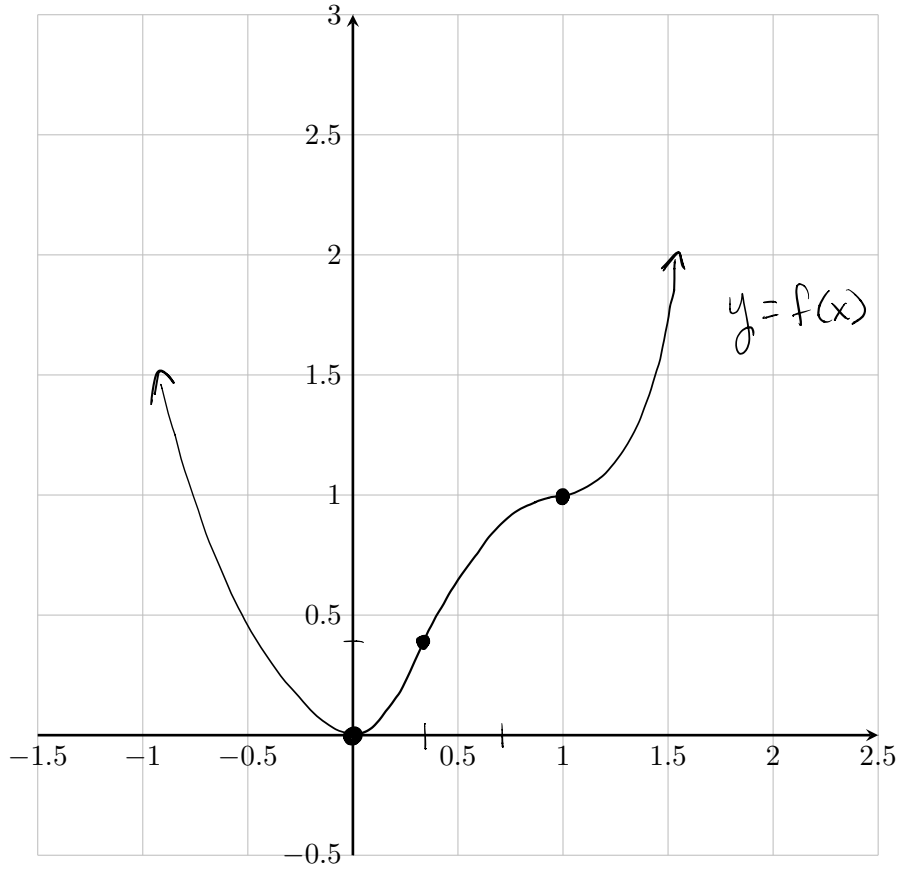
$f''(2) = 12(3(2)-1)(2-1) > 0$

Inflection Points occur at

$(1/3, f(1/3)) = (1/3, 11/27)$

$(1, f(1)) = (1, 1)$

(d) Plot the local extrema and the inflection points on the graph. Transfer the information from parts (a) and (b) to the number lines for $f'(x)$ and $f''(x)$. Sketch the graph of the function $f(x) = 3x^4 - 8x^3 + 6x^2$, using all of the information.



$f'(x)$: − +

$f''(x)$: + − +

(e) Now use your graphing calculator to get the graph of $y = f(x)$ on this domain, and compare it to the graph you just drew. How well did you do?

The actual graph (according to technology - remember, depending on your window the graph may look skewed) appears to be steeper (decrease and increase more rapidly) but overall, it is a good sketch!

2. Using the same process as in the previous problem, graph $f(x) = x^{\frac{1}{3}}(x+4)$ on the next page.

$$f(x) = x^{1/3}(x+4) = x^{4/3} + 4x^{1/3}$$

- x-intercepts at $x=0$, $x=-4$

- $f'(x) = \frac{4}{3}x^{1/3} + \frac{4}{3}x^{-2/3}$

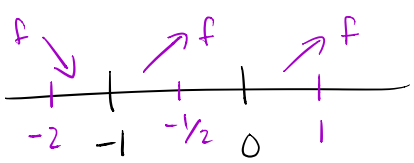
$$= \frac{4x^{1/3}}{3} + \frac{4}{3x^{2/3}}$$

$$= \frac{4x+4}{3x^{2/3}}$$

- Critical Points

$f'(x) = 0$ when $x = -1$

$f'(x)$ undefined when $x=0$



$f'(-2) < 0$

$f'(-1/2) > 0$

$f'(1) > 0$

- Increasing on $(-1,0) \cup (0,\infty)$
- Decreasing on $(-\infty,-1)$
- Local min at $x=-1$, $(-1,-3)$
- No local max

- $$f''(x) = \frac{(4x+4)'(3x^{2/3}) - (4x+4)(3x^{2/3})'}{(3x^{2/3})^2}$$

$$= \frac{4(3x^{2/3}) - (4x+4)(2x^{-1/3})}{9x^{4/3}}$$

$$= \frac{(12x^{2/3} - (4x+4)2x^{-1/3})}{9x^{4/3}} \cdot \frac{x^{1/3}}{x^{1/3}}$$

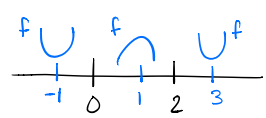
$$= \frac{12x - 2(4x+4)}{9x^{5/3}}$$

$$= \frac{12x-8x-8}{9x^{5/3}}$$

$$= \frac{4x-8}{9x^{5/3}}$$

- Inflection Points

$f''(x) = 0$ when $x=2$, $f''(x)$ undefined when $x=0$



$f''(-1) > 0$

$f''(1) < 0$

$f''(3) > 0$

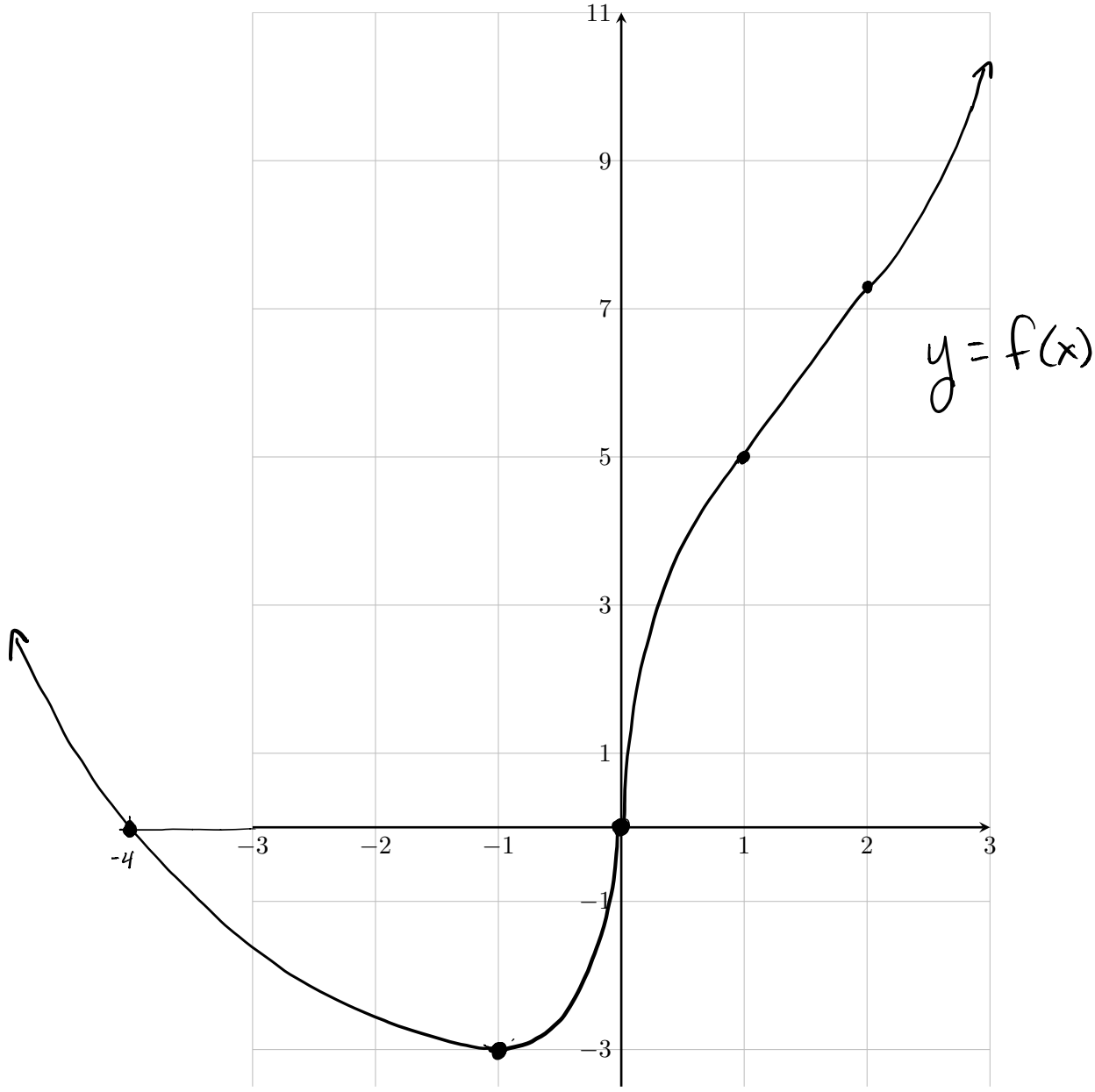
Inflection Points occur at

$x=0$, $(0,0)$

$x=2$, $(2, 6\sqrt[3]{2})$

- Concave up on $(-\infty,0) \cup (2,\infty)$
- Concave down on $(0,2)$

Graph of $f(x) = x^{\frac{1}{3}}(x+4)$



$f'(x)$: − ● + ○ +

$f''(x)$: + ○ − ● +